

Demir Guvel

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Education

Western University – BSc in Software Engineering, Class of 2028

Sep 2024 – Present

- **Relevant Coursework:** Data Structures & Algorithms | Software Construction | OOP | Software Design | Scripting | Databases | Web Technologies | Networking | Software Requirements | Software Project Management

Technical Skills

Languages: Python, Java, C++, C#, JavaScript, TypeScript, HTML, CSS

Frameworks & AI: React, Flask, Gemini API, YOLOv8-Pose, OpenCV

Tools: Git, GitHub, VS Code, IntelliJ IDEA, Unity, MATLAB, Quartus, ModelSim, Arduino IDE, Tailscale, Roboflow

Engineering: Onshape, SolidWorks, Tinkercad

Projects

Sentinel-AI – Real-Time AI Agent Monitoring System | Hack Canada 2026 | [Sentinel-AI](#) | [Devpost](#)

- Developed a real-time AI agent monitoring system using **Python**, **Gemini API**, and a custom weighted risk-scoring algorithm
- Implemented detection of loops, goal drift, low confidence, and tool incoherence with adaptive reprompt, rollback, decomposition, and halt interventions
- Built a **React/TypeScript** dashboard for live agent status, risk scores, and intervention history; implemented distributed communication using **Tailscale**

Dieta – AI Fitness & Nutrition Platform | GenAI Genesis 2026 | [Dieta](#) | [Devpost](#)

- Developed real-time exercise form analysis using **Python**, **Flask**, **YOLOv8-Pose**, and **OpenCV** from webcam video
- Calculated joint angles from 17 body keypoints to evaluate posture and applied temporal smoothing to handle limb occlusion
- Integrated **Gemini API** to generate personalized meal plans based on workout intensity and nutritional requirements

Power Monitor for Medical Devices | Western University | Jan. 2025 – Apr. 2025

- Developed a **Python** monitoring system integrating **Arduino** hardware to measure voltage and current for medical devices in low-resource environments
- Built a multilingual interface with **data visualization**, **logging**, battery monitoring, and warning alerts while meeting a \$100 project budget

Autonomous Turtle Egg Picker | Western Engineering Competition 2025

- Developed **C++** control software integrating sensors and **servo motors** to autonomously detect, navigate to, and collect simulated turtle eggs
- Implemented and tested **autonomous navigation** and task-execution logic as part of a multidisciplinary team

Engineering Experience & Activities

Fall Cohort Participant | IBM AI Experiential Learning Lab | Aug. 2026 – Present

- Developing an **AI solution** for a **Smart Healthcare** industry challenge using **IBM Bob** and **IBM Cloud**

Software Developer | Google Developer Student Club | Sep. 2025 – Present

- Participating in technical workshops and sessions covering **software development**, **APIs**, and **cloud technologies**

Leadership

Symphonic Band Section Leader, North Toronto Collegiate Institute | Jun. 2023 – Jun. 2024

- Led a **15-member section**, conducting 20+ weekly rehearsals and preparing for 5 concerts
- Contributed to a Silver Award at the OBA Provincial Band Festival while maintaining a **99% average** across two music courses

Interests

Classical Guitar | 2014 – Present | [Demir Guvel](#) | [YouTube](#)

- Performed in **10+** solo concerts for audiences of **500+** people, demonstrating discipline, confidence, and resilience
- Repertoire includes *Tango en Skaï*, *Recuerdos de la Alhambra*, *Asturias*, and *Capricho Árabe*